```
PRIVATE KEY d — VERTICAL STACK (solved puzzles from 53125.txt)

puzzles: 82  dec width: 40  hex width: 33
bit_length ~ puzzle number (d_bits - n is usually 0 or small)

=== per-puzzle d (decimal, right-aligned) ===

P  1  bits=  1  h2=  1 h3=   1 t3=  1                                         1
P  2  bits=  2  h2=  3 h3=   3 t3=  3                                         3
P  3  bits=  3  h2=  7 h3=   7 t3=  7                                         7
P  4  bits=  4  h2=  8 h3=   8 t3=  8                                         8
P  5  bits=  5  h2= 21 h3=  21 t3= 21                                        21
P  6  bits=  6  h2= 49 h3=  49 t3= 49                                        49
P  7  bits=  7  h2= 76 h3=  76 t3= 76                                        76
P  8  bits=  8  h2= 22 h3= 224 t3=224                                       224
P  9  bits=  9  h2= 46 h3= 467 t3=467                                       467
P 10  bits= 10  h2= 51 h3= 514 t3=514                                       514
P 11  bits= 11  h2= 11 h3= 115 t3=155                                      1155
P 12  bits= 12  h2= 26 h3= 268 t3=683                                      2683
P 13  bits= 13  h2= 52 h3= 521 t3=216                                      5216
P 14  bits= 14  h2= 10 h3= 105 t3=544                                     10544
P 15  bits= 15  h2= 26 h3= 268 t3=867                                     26867
P 16  bits= 16  h2= 51 h3= 515 t3=510                                     51510
P 17  bits= 17  h2= 95 h3= 958 t3=823                                     95823
P 18  bits= 18  h2= 19 h3= 198 t3=669                                    198669
P 19  bits= 19  h2= 35 h3= 357 t3=535                                    357535
P 20  bits= 20  h2= 86 h3= 863 t3=317                                    863317
P 21  bits= 21  h2= 18 h3= 181 t3=764                                   1811764
P 22  bits= 22  h2= 30 h3= 300 t3=503                                   3007503
P 23  bits= 23  h2= 55 h3= 559 t3=802                                   5598802
P 24  bits= 24  h2= 14 h3= 144 t3=676                                  14428676
P 25  bits= 25  h2= 33 h3= 331 t3=509                                  33185509
P 26  bits= 26  h2= 54 h3= 545 t3=862                                  54538862
P 27  bits= 27  h2= 11 h3= 111 t3=941                                 111949941
P 28  bits= 28  h2= 22 h3= 227 t3=408                                 227634408
P 29  bits= 29  h2= 40 h3= 400 t3=894                                 400708894
P 30  bits= 30  h2= 10 h3= 103 t3=084                                1033162084
P 31  bits= 31  h2= 21 h3= 210 t3=551                                2102388551
P 32  bits= 32  h2= 30 h3= 309 t3=814                                3093472814
P 33  bits= 33  h2= 71 h3= 713 t3=912                                7137437912
P 34  bits= 34  h2= 14 h3= 141 t3=157                               14133072157
P 35  bits= 35  h2= 20 h3= 201 t3=792                               20112871792
P 36  bits= 36  h2= 42 h3= 423 t3=980                               42387769980
P 37  bits= 37  h2= 10 h3= 100 t3=595                              100251560595
P 38  bits= 38  h2= 14 h3= 146 t3=592                              146971536592
P 39  bits= 39  h2= 32 h3= 323 t3=937                              323724968937
P 40  bits= 40  h2= 10 h3= 100 t3=950                             1003651412950
P 41  bits= 41  h2= 14 h3= 145 t3=147                             1458252205147
P 42  bits= 42  h2= 28 h3= 289 t3=463                             2895374552463
P 43  bits= 43  h2= 74 h3= 740 t3=825                             7409811047825
P 44  bits= 44  h2= 15 h3= 154 t3=071                            15404761757071
P 45  bits= 45  h2= 19 h3= 199 t3=597                            19996463086597
P 46  bits= 46  h2= 51 h3= 514 t3=612                            51408670348612
P 47  bits= 47  h2= 11 h3= 119 t3=170                           119666659114170
P 48  bits= 48  h2= 19 h3= 191 t3=443                           191206974700443
P 49  bits= 49  h2= 40 h3= 409 t3=525                           409118905032525
P 50  bits= 50  h2= 61 h3= 611 t3=764                           611140496167764
P 51  bits= 51  h2= 20 h3= 205 t3=876                          2058769515153876
P 52  bits= 52  h2= 42 h3= 421 t3=700                          4216495639600700
P 53  bits= 53  h2= 67 h3= 676 t3=124                          6763683971478124
P 54  bits= 54  h2= 99 h3= 997 t3=707                          9974455244496707
P 55  bits= 55  h2= 30 h3= 300 t3=460                         30045390491869460
P 56  bits= 56  h2= 44 h3= 442 t3=575                         44218742292676575
P 57  bits= 57  h2= 13 h3= 138 t3=492                        138245758910846492
P 58  bits= 58  h2= 19 h3= 199 t3=049                        199976667976342049
P 59  bits= 59  h2= 52 h3= 525 t3=191                        525070384258266191
P 60  bits= 60  h2= 11 h3= 113 t3=382                       1135041350219496382
P 61  bits= 61  h2= 14 h3= 142 t3=982                       1425787542618654982
P 62  bits= 62  h2= 39 h3= 390 t3=062                       3908372542507822062
P 63  bits= 63  h2= 89 h3= 899 t3=768                       8993229949524469768
P 64  bits= 64  h2= 17 h3= 177 t3=628                      17799667357578236628
P 65  bits= 65  h2= 30 h3= 305 t3=855                      30568377312064202855
P 66  bits= 66  h2= 46 h3= 463 t3=726                      46346217550346335726
P 67  bits= 67  h2= 13 h3= 132 t3=302                     132656943602386256302
P 68  bits= 68  h2= 21 h3= 219 t3=825                     219898266213316039825
P 69  bits= 69  h2= 29 h3= 297 t3=804                     297274491920375905804
P 70  bits= 70  h2= 97 h3= 970 t3=481                     970436974005023690481
P 75  bits= 75  h2= 22 h3= 225 t3=367                   22538323240989823823367
P 80  bits= 80  h2= 11 h3= 110 t3=456                 1105520030589234487939456
P 85  bits= 85  h2= 21 h3= 210 t3=920                21090315766411506144426920
P 90  bits= 90  h2= 86 h3= 868 t3=863               868012190417726402719548863
P 95  bits= 95  h2= 25 h3= 255 t3=212             25525831956644113617013748212
P100  bits=100  h2= 86 h3= 868 t3=142            868221233689326498340379183142
P105  bits=105  h2= 29 h3= 290 t3=435          29083230144918045706788529192435
P110  bits=110  h2= 10 h3= 109 t3=748        1090246098153987172547740458951748
P115  bits=115  h2= 31 h3= 314 t3=950       31464123230573852164273674364426950
P120  bits=120  h2= 91 h3= 919 t3=323      919343500840980333540511050618764323
P125  bits=125  h2= 37 h3= 376 t3=420    37650549717742544505774009877315221420
P130  bits=130  h2= 11 h3= 110 t3=577  1103873984953507439627945351144005829577

=== leading 8 decimal digits — vertical columns ===
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  head[0] P  1-P40                          13512412371241131
  head[0] P 41-P120  1271151146246934115113813412292128282139
  head[0] P125-P130  31
  head[1] P  1-P40                       13543412001014020420
  head[1] P 41-P120  4845911901027904392149970631972116569011
  head[1] P125-P130  71
  head[2] P  1-P40                    13880541517030931130630
  head[2] P 41-P120  5904949191516702895320975329705008580949
  head[2] P125-P130  60
  head[3] P  1-P40                12599561092839673237318297 3
  head[3] P 41-P120  8590906211863441290558396468243590228063
  head[3] P125-P130  53
  head[4] P  1-P40             125061587317885843013443275726
  head[4] P 41-P120  2384686014746458477073298659738501523244
  head[4] P125-P130  08
  head[5] P  1-P40          245162585865375865894868730871145
  head[5] P 41-P120  5717466680698537560487263268463232812413
  head[5] P125-P130  57
  head[6] P  1-P40       247261581461263160070694828277765591
  head[6] P 41-P120  2416676994953594763172967192492011323625
  head[6] P125-P130  43
  head[7] P  1-P40   1378196474536470395743269240905892196364
  head[7] P 41-P120  2501305709569202568355977746973059130030
  head[7] P125-P130  99

=== trailing 8 decimal digits — vertical columns ===
  tail[-7] P  1-P40                          13512412371241131
  tail[-7] P 41-P120  1271151146246934115113813412292128282139
  tail[-7] P125-P130  31
  tail[-6] P  1-P40                       13543412001014020420
  tail[-6] P 41-P120  4845911901027904392149970631972116569011
  tail[-6] P125-P130  71
  tail[-5] P  1-P40                    13880541517030931130630
  tail[-5] P 41-P120  5904949191516702895320975329705008580949
  tail[-5] P125-P130  60
  tail[-4] P  1-P40                125995610928396732373182973
  tail[-4] P 41-P120  8590906211863441290558396468243590228063
  tail[-4] P125-P130  53
  tail[-3] P  1-P40             125061587317885843013443275726
  tail[-3] P 41-P120  2384686014746458477073298659738501523244
  tail[-3] P125-P130  08
  tail[-2] P  1-P40          245162585865375865894868730871145
  tail[-2] P 41-P120  5717466680698537560487263268463232812413
  tail[-2] P125-P130  57
  tail[-1] P  1-P40       247261581461263160070694828277765591
  tail[-1] P 41-P120  2416676994953594763172967192492011323625
  tail[-1] P125-P130  43
  tail[+0] P  1-P40   1378196474536470395743269240905892196364
  tail[+0] P 41-P120  2501305709569202568355977746973059130030
  tail[+0] P125-P130  99

=== trailing 16 hex nibbles — vertical columns ===
  hex[-15] P  1-P40
  hex[-15] P 41-P120                      137f127b134e125a136b
  hex[-15] P125-P130  13
  hex[-14] P  1-P40
  hex[-14] P 41-P120                  127f36c7a83e04ca1c2f6501
  hex[-14] P125-P130  c3
  hex[-13] P  1-P40
  hex[-13] P 41-P120              1269ec4cc3c0830b19517e75fcf0
  hex[-13] P125-P130  5e
  hex[-12] P  1-P40
  hex[-12] P 41-P120          127e83adb6909de532fbdbca20a5104f
  hex[-12] P125-P130  37
  hex[-11] P  1-P40
  hex[-11] P 41-P120      126a725f06b1276765518ec388e5007f4dd2
  hex[-11] P125-P130  36
  hex[-10] P  1-P40
  hex[-10] P 41-P120  126e2ecd4b0a7fe855caa4efbd29341ccb9cf712
  hex[-10] P125-P130  b6
  hex[-9] P  1-P40                                       124e
  hex[-9] P 41-P120  5ab02cde1d7e8b1bcbb131f217342b164b25c215
  hex[-9] P125-P130  65
  hex[-8] P  1-P40                                   134972b9
  hex[-8] P 41-P120  32d2f166740186f698b87ed734507646f2b92357
  hex[-8] P125-P130  b7
  hex[-7] P  1-P40                               137ba4ad535a
  hex[-7] P 41-P120  823bc81d63a6ed9305824bab5fcc546d011c0472
  hex[-7] P125-P130  b0
  hex[-6] P  1-P40                           136d7dd89aee78fe
  hex[-6] P 41-P120  61b3a307bc1445ba727526c0021df38c13835d4c
  hex[-6] P125-P130  75
  hex[-5] P  1-P40                       125df4a9e94666d87284
  hex[-5] P 41-P120  9c2518bc02ac7a6c91c3f1c95b90b16c86334ff4
  hex[-5] P125-P130  f3
  hex[-4] P  1-P40                   135dbd5ca0c124f2c5225f39
  hex[-4] P 41-P120  a57a485e1e0bed7458a661f1b54f2aa1dac5c759
  hex[-4] P125-P130  05
  hex[-3] P  1-P40               126c7072ae6253365ceaa9106a03
  hex[-3] P 41-P120  c8c33d33590931efd9b74a612e2cb611547cddd7
  hex[-3] P125-P130  89
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  hex[-2] P  1-P40           124a4989684c54eae28c5d76811aac33
  hex[-2] P 41-P120  cd55c5cbf39e2f1f6a4b9b8263c1cc3b14f88eea
  hex[-2] P125-P130  0f
  hex[-1] P  1-P40       134ed08763f340953050e67e1642d1779ded
  hex[-1] P 41-P120  589804b945d3641d124b0e0d85147435b56e7be8
  hex[-1] P125-P130  40
  hex[+0] P  1-P40   137851c0323b0036fdf54f245e58e47e8d0c3096
  hex[+0] P 41-P120  bf1f54abd44cc34fc1fe6e846ea9ee6a8c4ce0e3
  hex[+0] P125-P130  e4

=== head2 / head3 / tail3 lane stats ===

-- by lane3 --
  lane3=0 (n=27): top head2 19(3), 46(2), 26(2), 11(2), 29(2)
  lane3=1 (n=28): top head2 51(3), 30(2), 21(2), 14(2), 10(2)
  lane3=2 (n=27): top head2 11(3), 21(2), 10(2), 30(2), 14(2)

-- by lane5 --
  lane5=0 (n=26): top tail3 950(2), 21(1), 514(1), 867(1), 317(1)
  lane5=1 (n=14): top tail3 1(1), 49(1), 155(1), 510(1), 764(1)
  lane5=2 (n=14): top tail3 3(1), 76(1), 683(1), 823(1), 503(1)
  lane5=3 (n=14): top tail3 825(2), 7(1), 224(1), 216(1), 669(1)
  lane5=4 (n=14): top tail3 8(1), 467(1), 544(1), 535(1), 676(1)

-- by lane3 --
  lane3=0 (n=27): top tail3 7(1), 49(1), 467(1), 683(1), 867(1)
  lane3=1 (n=28): top tail3 950(2), 1(1), 8(1), 76(1), 514(1)
  lane3=2 (n=27): top tail3 3(1), 21(1), 224(1), 155(1), 544(1)

=== d_bits - n (should be ~0) ===
  delta=+0: 82 puzzles

=== no d in 53125 (78 puzzles) ===
  P71, P72, P73, P74, P76, P77, P78, P79, P81, P82, P83, P84, P86, P87, P88, P89, P91, P92, P93, P94, P96, P97, P98, P99, P101, P1
  ... +18 more
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